

File Handling

1. What is File Handling

File Handling refers to the process of creating, reading, writing, appending, and managing files stored on secondary storage (like HDD/SSD) using a programming language.

Without file handling, data would only exist temporarily in memory (RAM) and disappear when the program stops.

```
#Why File Handling is Important?
# 1. Data Persistence
# 2. Logs
# 3. Configuration Management (Environment etc)
```

2. Types of Files

```
#1. Text Files: Text files store data in human-readable format. (.txt, .csv, .json)
#2. Binary files: store data in non-human-readable format (0s and 1s). (.bin, images, executables)
```

3. Opening and Closing Files in Python

```
# 1.open() Function: The open() function is used to open a file and return a file object.
```

```
#Syntax:
# open(file_name, mode)
# file_name → Name (or path) of the file
# mode → How you want to interact with the file
```

```
file = open(r"D:\data1.txt", "w")
```

```
# Write Mode: w
```

```
#If File Doesn't exist it will create it.
```

```
#If File exists, old contents are immediately erased.
```

```
file = open(r"D:\data1.txt", "w")  
file.write("Bye")  
file.close()
```

```
#Read Mode: r
```

```
# Opens file for reading
```

```
# File must exist
```

```
# Default mode
```

```
# If file does not exist → FileNotFoundError
```

```
file = open(r"D:\data1.txt", "r")  
content = file.read()  
file.close()  
print(content)
```

```
Bye
```

```
#Append Mode: a
```

```
# Adds new content at the end
```

```
# Does NOT delete existing data
```

```
# Creates file if it doesn't exist
```

```
file = open(r"D:\data2.txt", "a+")  
file.write("Line2")  
print(file.read())  
file.close()
```

```
file = open("data2.txt", "r")  
content = file.read()  
file.close()  
print(content)
```

```
#Create Mode: x
# Creates a new file
# If file already exists → ❌ Error
# Useful when you want to prevent overwriting.
```

```
file = open(r"D:\data2.txt", "x")
```

```
FileExistsError: [Errno 17] File exists: 'D:\\data2.txt'
```

```
##'r+'
##Read + Write
##File must exist
```

```
##'w+'
##Write + Read
##Overwrites file
```

```
##'a+'
##Append + Read
##Adds content and allows reading
```

Context Manager: with

#The with statement is used to handle resources (like files) automatically and safely.

```
with open(r"D:\data2.txt", "r") as file:
    content = file.read()
    print(content)
```

```
#File is opened
#Block of code is executed
#File is automatically closed (even if an error occurs)
```

```
Line1Line2Line2Line2Line2
```

```
with open("data.txt", "r") as file:
    print(file.read())
    x = 10 / 0    # Error
```

```
#read() - Read Entire File
```

```
#The read() method reads the entire content of the file as a single string.
```

```
with open(r"D:\data2.txt", "r") as file:  
    content = file.read()  
    print(content)  
#Loads entire file into memory (RAM)  
#Not suitable for very large files
```

```
Line1Line2Line2Line2Line2Line2
```

```
#read specific no of characters  
with open(r"D:\data2.txt", "r") as file:  
    content = file.read(8)  
    print(content)
```

```
Line1Lin
```

```
#Read one line at a time  
with open("data.txt", "r") as file:  
    line1 = file.readline()  
    line2 = file.readline()  
    print(line1)  
    print(line2)
```

```
Hello World
```

```
New Line
```

```
#read all lines in a list  
with open("data.txt", "r") as file:  
    lines = file.readlines()  
    print(lines)
```

```
['Hello World\n', 'New Line\n', 'New Line']
```

```
#For Large files
with open("data.txt", "r") as f:
    for line in f:
        print(line.strip())
```

```
Hello World
New Line
New Line
```

```
# Example 1: Using write() to write strings to a file

# Step 1: Open file in write mode and write some strings
with open("example_write.txt", "w") as file:
    file.write("Hello World!\n")
    file.write("This is a file written using write() method.\n")
    file.write("Python File Handling is easy!\n")

# Step 2: Open the same file in read mode and read all lines
with open("example_write.txt", "r") as file:
    content = file.read()
    print("Content of example_write.txt:")
    print(content)
```

```
Content of example_write.txt:
Hello World!
This is a file written using write() method.
Python File Handling is easy!
```

```
# Example 2: Using writelines() to write a list of strings to a file

# List of strings (each string must end with '\n' for new line)
lines = [
    "Line 1: This is the first line.\n",
    "Line 2: Using writelines() method.\n",
    "Line 3: Each item in the list becomes a line.\n"
]

# Step 1: Open file in write mode and write the list of strings
with open("example_writelines.txt", "w") as file:
    file.writelines(lines)

# Step 2: Open the same file in read mode and read all lines
with open("example_writelines.txt", "r") as file:
    content = file.read()
    print("Content of example_writelines.txt:")
    print(content)
```

```
Content of example_writelines.txt:
Line 1: This is the first line.
Line 2: Using writelines() method.
Line 3: Each item in the list becomes a line.
```